

Jaume **Ivars Grimalt**

AI Engineer

Currently in Valencia, Spain | EU citizen | Open to relocation to Zurich, Munich, London

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Profile

I take models from the idea to production: data, training, deployment, backend and cloud. At Mycrospace I was the only engineer and built the whole stack, from a Master's thesis to a computer vision product used by 700 users in 35 countries.

Technical Skills

ML/DL	PyTorch (DDP, AMP, torch.compile), TensorFlow/TFLite, Hugging Face, timm, scikit-learn, OpenCV, pandas, ONNX
MLOps & cloud	GCP (Cloud Run, Cloud SQL, GCS), Hetzner/Dokploy, Docker, Redis/BullMQ, CI/CD, SLURM
Backend & data	Python, Flask, NestJS, PostgreSQL/Prisma, MongoDB
Generative AI	LLMs via API, agents, diffusion (LDM, ControlNet, IP-Adapter)

Experience

Mycrospace — *Valencia, Spain*

Co-Founder & CTO — *Jan 2024 – Oct 2026*

- Computer vision product for microbiology labs. Master's thesis (2024) to company (Jan 2026). Sole engineer; built the whole stack. **700** users in 35 countries; tested in **5 labs**. 1st Prize IdeasUPV (2024).
- **Production.** Event-based image pipeline: BullMQ/Redis queues, a Python Flask inference service, signed-URL uploads and SSE/WebSocket. Docker on GCP (Cloud Run, Cloud SQL, GCS), migrated to Hetzner/Dokploy to cut costs. NestJS/Prisma/PostgreSQL, MongoDB, Next.js, React Native/Expo and Stripe.
- **MLOps. 1,500**/**5,000** images annotated with microbiologists. Annotation versioning in MongoDB, k-fold CV, WBF ensembles, Cleanlab label audits and retraining on real client data. Vision models quantized and compiled with TFLite for edge.
- **Models.** Own ConvNeXt detector (YOLO and DQ-DETR-inspired) for small objects and dense scenes; 10–2000 colonies/image. **67 mAP50-95**; 92% recall / 93% precision. **12,500+** GPU-hours on Leonardo (two EuroHPC AI Factory allocations), **4x A100**, DDP/AMP/torch.compile on SLURM, checkpoint-resumed jobs.

MIALAB: Medical Imaging Analysis Laboratory, UPV — *Valencia, Spain*

Research Engineer (full-time) — *May 2025 – Jan 2026*

- Volumetric brain MRI segmentation with nnU-Net. Profiled volBrain (**700k+** volumes) from NIfTI ingest to voxel prediction for inference cost.

Tapstar — *Valencia, Spain*

Full-Stack Engineer (freelance) — *Jan 2025 – May 2025*

- Client delivery: QR generation, a back office and the frontend.

Vesta-Z — *Valencia, Spain*

Data Engineer — *Jul 2023 – Feb 2025*

- Cloud product features and a custom database schema so items could be cross-matched across providers. Delivered an LLM customer-support bot.

Earlier experience: Neurocatching, Data Scientist Intern, 2023; Sciling, Backend Developer Intern, 2022.

Generative AI

- Python pipeline of LLM-based agents that automates profile research and drafts outreach emails and messages.
- Prototype, not deployed: synthetic data augmentation with diffusion (LDM, ControlNet and IP-Adapter).

Education

M.Sc. Artificial Intelligence, Pattern Recognition & Digital Imaging — *Valencia, Spain*

UPV · GPA 8.7/10 · Thesis Honours (became Mycrospace) — *2023 – 2024*

Technology Management — *Valencia / Munich*

CDTM (TUM/LMU) · 16 selected — *2024*

B.Sc. Computer Engineering — *Valencia, Spain*

UPV · GPA 8.3/10 · Honours in ML and Statistics — *2019 – 2023*

Teaching: Big Data module, CEI, Valencia, Oct 2026 – Dec 2026.

Publications

- *Auto-N2N: Self-supervised MRI Denoising using Synthetic Pair Generation and Mixture of Experts.* S. Morell-Ortega et al., incl. **J. Ivars Grimalt**. Preprint, 2026. OpenReview
- *Detecting Cytopathic Effect in Cell Culture with Computer Vision.* W. Ragazzo, **J. Ivars Grimalt**. Under review.
- *What Extra Euler Steps Reveal About Residual Networks: Feature Dynamics in Weight-Tied and Untied Stages.* Manuscript in preparation.

Languages: Spanish (native), Catalan (native), English (C1)